

Stephanie M. Lukin, Ph.D.

<https://slukin.github.io/>

<https://www.linkedin.com/in/stephanie-m-lukin>

stephanie.m.lukin.civ@army.mil

PROFESSIONAL EXPERIENCE

- 12/2023 – Present *Senior Computer Scientist*
U.S. DEVCOM Army Research Laboratory, Los Angeles, CA
- 12/2017 – 12/2023 *Computer Scientist*
U.S. DEVCOM Army Research Laboratory, Los Angeles, CA
- 5/2017 – 12/2017 *Postdoctoral Fellow*
U.S. DEVCOM Army Research Laboratory, Los Angeles, CA
- 6/2016 – 9/2016 *Software Engineering Intern*
Google, New York, NY
- 6/2015 – 9/2015 *Research Intern*
Microsoft Research, Redmond, WA
- 6/2013 – 9/2023 *Research Intern*
PARC, A Xerox Company, Palo Alto, CA

EDUCATION

- 2017 Ph.D. Computer Science, University of California, Santa Cruz
[Generating Variations in a Virtual Storyteller](#)
- 2013 M.S. Computer Science, University of California, Santa Cruz
- 2011 B.S. Computer Science, Loyola University Maryland, Magna Cum Laude.

RESEARCH INTERESTS

- Natural language processing and generation
- Narrative intelligence, visual storytelling
- Multi-modal human-robot dialogue

PUBLICATIONS

2026

- 🏆 Lukin, Stephanie M., Kimberly A. Pollard, Claire N. Bonial, Cory J. Hayes, Kallirroi Georgila, Ron Arstein, David Traum. [Non-Event Oriented Video Assessments in Long-Form, Robot Videos](#). Workshop on Multimodal Augmented Generation via Multimodal Retrieval at ACL, 2026. **Best Paper Award.**
- Sahetai, Muskan, Cynthia Matuszek, An May, Francis Ferraro, Kasra Darvish, Stephanie Lukin, Ruchen Wen, Evie Newell. [What and Why: Benchmarking Bidirectional Intentional Reasoning in Multimodal Robot Tasks](#). IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), 2026.
- Ogbadu, Ekele, Stephanie Lukin, Cynthia Matuszek. [Ask-to-Act: Learning When Robots Should Clarify Ambiguous Grounded Instructions](#). IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), 2026.
- Indukuri, Murali Krishna Varma, Mohammad Eskandari, Sree Nitya Kollu, Stephanie Lukin, Cynthia Matuszek. [Hazard or Anomaly? Evaluating VLMs for Understanding Dangers and Discrepancies](#). IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), 2026.
- Eskandari, Mohammad, Murali Krishna Varma Indukuri, Sree Nitya Kollu, Stephanie Lukin, Cynthia Matuszek. [Autonomous VR-Based Risk Detection for Situational Awareness in Dangerous Settings](#). IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), 2026.

2025

- Chang, Aiden, Celso de Melo, and Stephanie M. Lukin. [Aha! - Predicting What Matters Next: Online Highlight Detection Without Looking Ahead](#). Conference on Neural Information Processing Systems (NeurIPS), 2025.
- Halperin, Brett and Stephanie M. Lukin. [From Camera-Eye to AI: Exploring the Interplay of Cinematography and Computational Visual Storytelling](#). ACM (Association of Computing Machinery) CHI conference on Human Factors in Computing Systems, 2025.
- Bowser, Shawn, Cynthia Matuszek, and Stephanie M. Lukin. [Towards Integrated Multimodal Interaction: Merging Immersive 3D Worlds with Language Based Retrieval for 3D Scene Understanding](#). Intelligent Cross-Data Analysis and Retrieval Workshop at ICMR, 2025.
- Eskandari, Mohammad, Murali Indukuri, Stephanie Lukin, and Cynthia Matuszek. [LLM-Supported Safety Annotation in High-Risk Environment](#). International Workshop on Virtual, Augmented, and Mixed Reality for Human-Robot Interaction at HRI, 2025.
- Higgins, Padraig, Cory Hayes, Stephanie M. Lukin, Cynthia Matuszek. [Language and Gesture in Virtual Reality: Is a Gesture Worth 1000 Words?](#) AAAI Fall Symposium: Unifying Representations for Robot Application Development, 2025.
- Ogbadu, Ekele, Stephanie Lukin, and Cynthia Matuszek. [Grounded Instruction Understanding with Large Language Models: Toward Trustworthy Human-Robot Interaction](#). AAAI Fall Symposium: AI Trustworthiness and Risk Assessment for Challenging Contexts, 2025.

2024

- Lukin, Stephanie M., Claire N. Bonial, Matthew Marge, Taylor Hudson, Cory J. Hayes, Kimberly A. Pollard, Anthony Baker, Ashley Foots, Ron Artstein, Felix Gervits, Mitchell Abrams, Cassidy Henry, Lucia Donatelli, Anton Leuski, Susan G. Hill, David Traum, and Clare Voss. [SCOUT: A Situated and Multi-Modal Human-Robot Dialogue Corpus](#). The Joint Language Resources and Evaluation Conference and Computational Linguistics Conference (LREC-COLING), 2024.
- Lukin, Stephanie M., Jaci South, and Shawn Bowser. [CHRIS-Bot: A Robot for Dialogue and Scene Understanding of Anomalous Environments in Virtual Reality](#). DEVCOM ARL Technical Report, ARL-TR-9906. 2024.
- Lukin, Stephanie M., Shawn Bowser, Reece Suchocki, Doug Summers-Stay, Jeffrey Micher, Cynthia Matuszek, Francis Ferraro, and Clare Voss. [MUMOSA, Interactive Dashboard for Multi-MOdal Situation Awareness](#). Future of Event Detection Workshop at EMNLP, 2024.
- Bonial*, Claire, Lukin*, Stephanie M., Mitchell Abrams, Anthony Baker, Lucia Donatelli, Ashley Foots, Cory J. Hayes, Cassidy Henry, Taylor Hudson, Matthew Marge, Kimberly A. Pollard, Ron Artstein, David Traum, Clare R. Voss. [Human-Robot Dialogue Annotation for Multi-Modal Common Ground](#). Journal on Language Resources and Evaluation (LREV), 2024
- 🏆 Bowser, Shawn and Stephanie M. Lukin. [3D Gaussian Splatting for Human-Robot Interaction](#). Interactive AI for Human-Centered Robotics Workshop at RO-MAN, 2024. **Best Presentation Award**.
- Halperin, Brett and Stephanie M. Lukin. [Artificial Dreams: Surreal Visual Storytelling as Inquiry Into AI 'Hallucination'](#). ACM Designing Interactive Systems (DIS), 2024.
- Higgins, Padraig, Ryan Barron, Stephanie Lukin, Don Engel, and Cynthia Matuszek. [A Collaborative Building Task in VR vs. Reality](#). *International Symposium on Experimental Robotics (ISER)*, 2024.

2023

- Lukin, Stephanie M., Claire Bonial, Taylor Hudson, Kimberly A. Pollard, Ron Artstein, Clare Voss, and David Traum. [Navigating to Success in Multi-Modal Human-Robot Collaboration](#). *IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, 2023.
- Lukin, Stephanie M., and Sungmin Eum. [SEE&TELL: Controllable Narrative Generation from Images](#). AAAI Creative AI Across Modalities Workshop, 2023.
- Lukin, Stephanie M. and Rahul Sharma. [Anomaly Detection with Visual Question Answering](#). DEVCOM Tech Report ARL-TR-9817, 2023.
- Lukin, Stephanie, Rahul Sharma, Michael Bellissimo. [Learning to Understand Anomalous Scenes from Human Interactions](#). DEVCOM Tech Report ARL-TR-9624, 2023.
- Halperin, Brett and Stephanie M. Lukin. [Envisioning Narrative Intelligence: Crafting a Creative Visual Storytelling Anthology](#). *CHI Human-Computer Interaction*, 2023.

- Summers-Stay, Doug, Clare R. Voss, Stephanie M. Lukin. [Brainstorm, then select: a generative language model improves its creativity score](#). *AAAI Creative AI Across Modalities Workshop*, 2023.
 - Marge, Matthew, Claire Bonial, Stephanie Lukin, Clare Voss. [Bot Language \(Summary Technical Report, Oct 2016-Sep 2021\)](#). March 2023, ARL-STR-9656.
- 2022
- Bonial, Claire, Taylor A Hudson, Austin Blodgett, Stephanie M. Lukin, Jeffrey Micher, Douglas Summers-Stay, Peter Sutor, Clare R Voss. [You Can't Quarantine the Truth: Lessons Learned in Logical Fallacy Annotation of an Infodemic](#). *International Conference on Language Resources and Evaluation (LREC)*, 2022.
 - Bonial, Claire, Taylor A Hudson, Austin Blodgett, Stephanie M. Lukin, Jeffrey Micher, Douglas Summers-Stay, Peter Sutor, Clare R Voss. [You Can't Quarantine the Truth: Lessons Learned in Logical Fallacy Annotation of an Infodemic](#). January 2022, ARL-TR-9343.
 - Bonial, Claire, Taylor A Hudson, Anthony Baker, Stephanie M. Lukin, David Traum. [Making Sense of Stop. Annotation, Recognition and Evaluation of Actions \(AREA II\) Workshop at ESSLI](#), 2022.
- 2021
- Lukin, Stephanie M., Clare R. Voss. [Extending Generation and Evaluation and Metrics \(GEM\) to Grounded Natural Language Generation \(NLG\) Systems and Evaluating their Descriptive Texts Derived from Image Sequences](#). ARL Technical Report, 2021.
 - Bonial, Claire, Mitchell Abrams, Anthony Baker, Taylor A. Hudson, Stephanie Lukin, David Traum, Clare R. Voss. [Context Is Key: Annotating Situated Dialogue Relations in Multi-floor Dialogue](#). Workshop on the Semantics and Pragmatics of Dialogue (SemDial), 2021.
- 2020
- Bonial, Claire, Lucia Donatelli, Mitchell Abrams, Stephanie M. Lukin, Stephen Tratz, Matthew Marge, Ron Artstein, David Traum, and Clare Voss. [Dialogue-AMR: Abstract Meaning Representation for Dialogue](#). *International Conference on Language Resources and Evaluation (LREC)*, 2020.
 - Sapkale, Aishwarya, and Stephanie M. Lukin. [Maintaining Consistency and Relevancy in Multi-Image Visual Storytelling](#). *ARL Tech Report*, 2020.
 - Bonial, Claire, Stephanie M. Lukin, David C. Doughty, Steven C. Hill, and Clare R. Voss. [InfoForager: Leveraging Semantic Search with AMR for COVID-19 Research](#). *Designing Meaning Representations Workshop (DMR)*, 2020.
- 2019
- Lukin, Stephanie M. and Marilyn Walker. [A narrative sentence planner and structurer for domain independent, parameterizable storytelling](#). *Dialogue and Discourse* 10 (1), 2019.
 - Bonial, Claire, Lucia Donatelli, Stephanie M. Lukin, Stephen Tratz, Ron Artstein, David Traum, and Clare Voss. [Augmenting Abstract Meaning Representation for Human-Robot Dialogue](#). *Designing Meaning Representations Workshop (DMR)*, 2019.
 - Marge, Matthew, Stephen Nogar, Cory J Hayes, Stephanie M. Lukin, Jesse Bloecker, Eric Holder, and Clare Voss. [A Research Platform for Multi-Robot Dialogue with Humans](#). *North American Association for Computational Linguistics Demo Paper (NAACL)*, 2019.
 - Bonial, Claire, David Traum, Traum Henry, Stephanie M Lukin, Matthew Marge, Ron Artstein, Kimberly A Pollard, Ashley Foots, Anthony L. Baker, and Clare R. Voss. [Dialogue Structure Annotation Guidelines for Army Research Laboratory \(ARL\) Human-Robot Dialogue Corpus](#). *ARL Tech Report*, 2019.
- 2018
- Marge, Matthew, Claire Bonial, Stephanie M. Lukin, Cory J. Hayes, Ashley Foots, Ron Artstein, Cassidy Henry, Kimberly A. Pollard, Carla Gordon, Felix Gervits, Anton Leuski, Susan G. Hill, Clare R. Voss, and David Traum. [Balancing Efficiency and Coverage in Human-Robot Dialogue Collection](#). *AI-HRI AAAI-FSS*, 2018.
 - Lukin, Stephanie M., Kimberly A. Pollard, Claire Bonial, Matthew Marge, Cassidy Henry, Ron Artstein, David Traum, and Clare R. Voss. [Consequences and Factors of Stylistic Differences in Human-Robot Dialogue](#). *SIGdial Meeting on Discourse and Dialogue (SIGDIAL)*, 2018.
 - Oraby, Shereen, Lena Reed, Shubhanghi Tandon, Sharath T.S., Stephanie Lukin, and Marilyn Walker. [Controlling Personality-Based Stylistic Variation with Neural Natural Language Generators](#). *SIGdial Meeting on Discourse and Dialogue (SIGDIAL)*, 2018.

- Lukin, Stephanie M., Felix Gervits, Cory Hayes, Pooja Moolchandani, Anton Leuski, John Rogers, Carlos Sanchez Amaro, Matthew Marge, Clare Voss and David Traum. [ScoutBot: A Dialogue System for Collaborative Navigation](#). *Association for Computational Linguistics Demo Paper (ACL)*, 2018.
 - Lukin, Stephanie M., Reginald Hobbs, Clare R. Voss. [A Pipeline for Creative Visual Storytelling](#). *Storytelling Workshop (StoryNLP) at NAACL*, 2018
 - Pollard, Kimberly A., Stephanie M. Lukin, Susan Hill, Ashley Foots, Matthew Marge. [How We Talk with Robot: Eliciting Minimally-Constrained Speech to Build Natural Language Interfaces](#). *Human Factors and Ergonomics Society (HFES)*, 2018.
 - Bonial, Claire, Stephanie Lukin, Ashley Foots, Cassidy Henry, Matthew Marge, Kimberly Pollard, Ron Artstein, David Traum, and Clare R. Voss. [Human-Robot Dialogue and Collaboration in Search and Navigation](#). *Annotation, Recognition and Evaluation of Actions (AREA) Workshop at LREC*, 2018.
 - Traum, David, Cassidy Henry, Stephanie Lukin, Ron Artstein, Felix Gervits, Kimberly Pollard, Claire Bonial, Su Lei, Clare Voss, Matthew Marge, Cory Hayes and Susan Hill. [Dialogue Structure Annotation for Multi-Floor Interaction](#). *International Conference on Language Resources and Evaluation (LREC)*, 2018.
- 2017
- Bonial, Claire, Matthew Marge, Ron Artstein, Ashley Foots, Felix Gervits, Cory J. Hayes, Cassidy Henry, Susan G. Hill, Anton Leuski, Stephanie M. Lukin, Pooja Moolchandani, Kimberly A. Pollard, David Traum, Clare R. Voss. [Laying Down the Yellow Brick Road: Development of a Wizard-of-Oz Interface for Collecting Human-Robot Dialogue](#). *Symposium on Natural Communication for Human-Robot Collaboration, AAAI Fall Symposium Series*, 2017.
 - Bowden, Kevin, Shereen Oraby, Amita Misra, Jiaqu Wu, and Stephanie M. Lukin. [Data-Driven Dialogue Systems for Social Agents](#). *International Workshop on Spoken Dialogue Systems position paper (IWSDS)*, 2017.
 - Lukin, Stephanie M., Pranav Anand, Marilyn Walker, Steve Whittaker. [Argument Strength is in the Eye of the Beholder: Audience Effects in Persuasion](#). *European Chapter of the Association for Computational Linguistics (EACL)*, 2017.
- 2016
- Lukin, Stephanie M., Kevin Bowden, Casey Barackman, and Marilyn A. Walker. [A Corpus of Personal Narratives and Their Story Intention Graphs](#). *International Conference on Language Resources and Evaluation (LREC)*, 2016.
- 2015
- Lukin, Stephanie M., Lena I. Reed, and Marilyn A. Walker. [Generating Sentence Planning Variations for Story Telling](#). *SIGdial Meeting on Discourse and Dialogue (SIGDIAL)*, 2015.
 - Lukin, Stephanie M. and Marilyn A. Walker. [Narrative Variations with a Virtual Storyteller](#). *Intelligent Virtual Agents (IVA)*, 2015.
- 2014
- Lukin, Stephanie M., James O. Ryan, and Marilyn A. Walker. [Automating Direct Speech Variations in Stories and Games](#). *Workshop on Games and NLP (GAMNLP)*, 2014.
 - Lukin, Stephanie M., Luke Eisenberg, Thomas Corcoran, and Marilyn A. Walker. [Identifying Subjective and Figurative Language in Online Dialogue](#). *Pacific Northwest Regional NLP Workshop (NWNLP)*, 2014.
 - Lukin, Stephanie M., G.M. Youngblood, Honglu Du, and Marilyn A. Walker. [Building Community & Commitment with a Virtual Coach in Mobile Wellness Programs](#). *Intelligent Virtual Agents (IVA)*, 2014.
 - Swanson, Reid, Stephanie Lukin, Luke Eisenberg, Thomas Corcoran, and Marilyn A. Walker. [Getting Reliable Annotations for Sarcasm in Online Dialogues](#). *International Conference on Language Resources and Evaluation (LREC)*, 2014.
 - Justo, Raquel, Thomas Corcoran, Stephanie Lukin, Marilyn A. Walker, M. Ines Torres. [Extracting Relevant Knowledge for the Detection of Sarcasm and Nastiness in the Social Web](#). *Knowledge-Based Systems (KBS)*, 2014.
- 2013
- Lukin, Stephanie, and Marilyn Walker. [Really? Well. Apparently Bootstrapping Improves the Performance of Sarcasm and Nastiness Classifiers for Online Dialogue](#). *Workshop on Language Analysis in Social Media (LASM) at NAACL*, 2013.

- Rishes, Elena, Stephanie M. Lukin, David K. Elson, and Marilyn A. Walker. [Generating Different Story Tellings from Semantic Representations of Narrative](#). *International Conference on Interactive Digital Storytelling (ICIDS)*, 2013.

INVITED TALKS

- 2020 Keynote Speaker | "How Many Different Ways Could You Tell a Story about a Sequence of Images? Procedural Expression for Narrative and Dialogue Generation and Visual Storytelling" at [The 16th AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment \(AIIDE'20\)](#)
- 2020 Keynote Speaker | "Where to next? Situated Dialogue with Robots for Search and Navigation." at [Semantics and Pragmatics of Dialogue \(SemDial'20\)](#)

PRIOR RESEARCH EXPERIENCE

- 4/2013 – 03/2017 Dissertation: Generating Variations in a Virtual Storyteller
Graduate Student Researcher, UC Santa Cruz, CA
 - Developed algorithms for mapping semantic to syntactic structures
 - Built a content planner and sentence planner to employ narrative devices for changing focalizing agent and adding emotions in character speech
Overgenerate and rank to evaluate engagement through personalization in stories and games
- 6/2016 – 9/2016 Dialogue Management
Software Engineering Intern, Google, New York, NY
 - Created a domain independent dialog manager
 - Collaborated with researchers to design a domain independent grammar
 - Built a sustainable and reusable pipeline for scraping training data
- 6/2015 – 9/2015 Identifying Emotions in Online Conversations
Summer Research Intern, Microsoft Research, Redmond, WA
 - Curated and computed statistics of a crowdsourced corpus of 12K conversations with sentiment and emotion annotations
 - Manipulated recurrent neural network architecture to generate personalized responses
- 5/2012 – 5/2014 Master's Thesis: Natural Language Processing for Sarcasm Detection
Graduate Student Researcher, UC Santa Cruz, CA
 - Built cue-based sarcasm classifier in socio-political debates
Improved precision by 40% over baseline methods by learning sarcasm specific syntactic patterns combined with cue words
- 6/2013 – 9/2013 Virtual Conversational Agents for iPhone Health and Fitness App
Summer Research Intern, PARC, A Xerox Company, Palo Alto, CA
 - Worked with the Fittle™ iOS wellness app team to build and pilot a fitness routine
 - Implemented a virtual coach to encourages community and delivers personalized attention
 - Increasing compliance and sense of community with virtual coach

- 1/2012 – 10/2012 Machine Learning on Factual and Emotional Political Arguments
Graduate Student Researcher, UC Santa Cruz, CA
- Built models to predict the degree of opinion change as a function of personality and argument type of dialogues in socio-political debates
 - Observed opinion change was greater for people initially neutral about a topic than those who were entrenched
 - Naïve Bayes and linear regression results from Weka (opinion change measured as binary and numeric) indicate the importance of personalized arguments adapted to people's personality
 -
- 6/2009 – 5/2010 Natural Language Processing for Identifying Malware in Source Code
Hauber Science Research Fellowship, Loyola University MD, Baltimore, MD
- Created models for different authors of code based on author differences in naming and commenting conventions
 Correctly identified an unlabeled document from a collection of code written by distinct authors 90% of time using Kullback-Liebr divergence

PROGRAM COMMITTEES

2026	Reviewer Language Resources and Evaluation (LREC'26), Student Research Workshop at EACL (SRW'26) International Conference on Intelligent Robots and Systems (IROS'26) Conversational User Interfaces (CUI'26) International Conference on Robot and Human Interactive Communication (RO-MAN'26)
2025	Reviewer Computer Human Interaction (CHI'25), Empirical Methods for Natural Language Processing (EMNLP'25), International Conference on Robotics and Automation (ICRA'25), Nations of America Association for Computational Linguistics (NAACL'25), Workshop on SPACE in Vision, Language, and Embodied AI (SpaVLE'25)
2024	OC Games and Natural Language Processing (Games and NLP'24) at LREC-COLING 2024 Reviewer Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING'24), IEEE Robotics and Automation Letters Journal (RA-L), Workshop on Spatial Language Understanding and Grounded Communication for Robotics (SpLU-RoboNLP'24)
2023	Reviewer Association for Computational Linguistics (ACL'23)
2022	OC Games and Natural Language Processing (Games and NLP'22) at LREC 2022 Area Chair International Conference on Computational Linguistics (COLING'22) Reviewer International Conference on Interactive Digital Storytelling (ICIDS'22), ACL Rolling Review (ARR'22), Special Interest Group on Dialogue (SIGDIAL'22), Creative AI Across Modalities, Conference on Autonomous Agents and Multiagent Systems (AAMAS'22)
2021	Reviewer ACL Rolling Review (ARR'21)
2020	Co-Chair Games and Natural Language Processing (GAMNLP'20) at LREC 2020 Co-Chair Special Session on Situated Dialogue with Virtual Agents and Robots (RoboDial'20) Session Chair Semantics and Pragmatics of Dialogue (SemDial'20) Reviewer Association for Computational Linguistics (ACL'20), Human-Robot Interaction (HRI'20), Language Resources and Evaluation (LREC'20), International Conference on Interactive Digital Storytelling (ICIDS'20)

2019	Co-Chair Storytelling Workshop (StoryNLP'19) at ACL 2019 Co-Chair Games and Natural Language Processing (GAMNLP'19) at FDG 2019 Publication Co-Chair North American Association Computational Linguistics (NAACL'19) Area Chair, Generation Association for Computational Linguistics (ACL'19) Reviewer North American Association for Computational Linguistics (NAACL'19), Conference on Autonomous Agents and Multiagent Systems (AAMAS'19), Conference on Robot Learning (CoRL'19)
2018	Co-Chair Special Session on Physically Situated Dialogue (RoboDial'18) at SIGDIAL 2018 Publication Co-Chair North American Association Computational Linguistics (NAACL'18) Reviewer Association for Computational Linguistics (ACL'18) North American Association for Computational Linguistics (NAACL'18), Storytelling Workshop at NAACL (StoryNLP'18) , Student Research Workshop at NAACL (SRW'18), Joint workshop of Intelligent Narrative Technologies (INT'18) and Intelligent Cinematography and Editing (WICED'18)
2017	Reviewer International Conference on Interactive Digital Storytelling (ICIDS'17), Workshop on Intelligent Narrative Technologies (INT'17), International Joint Conference on Natural Language Processing (IJCNLP'17)
2016	Reviewer International Conference on Interactive Digital Storytelling (ICIDS'16)